

2.6 “All” and “Not” Statements

Lesson Introduction

	Benchmark	MA.912.LT.4.3 Identify and accurately interpret “if...then,” “if and only if,” “all” and “not” statements. Find the converse, inverse, and contrapositive of a statement.			Learning Targets	- I can write and interpret an “all” statement. - I can write and interpret a negation.
	Benchmark Coherence	Building On		Working Toward		
		N/A		N/A		
	Essential Questions	- How can I know if an “all” statement is true or not? - What is the relationship between “all” and “not” statements?				
	Lesson Narrative	In this lesson, students will learn about “all” and “not” statements. Students examine the truth value of statements using the word “all” and edit them to use the word “some” if the truth value was false. They will explore how Venn diagrams can be used to represent mathematical or logical sets pictorially when comparing all and some. Students will also be introduced to the concept of the negation of a statement.				
	Component Summary	2.6.1 Warm-Up (~5 min.)	Bell Work on converse, inverse, contrapositive, and biconditional (<i>SE p. 107</i>)	2.6.3 Guided Practice (10 min.)	Determining truth values of “all” statements and rewriting statements as conditionals and negations (<i>SE p. 110</i>)	
		2.6.2 Guided Instruction (30 min.)	Examining “all” and “not” statements (<i>SE p. 107</i>)	2.6.4 Wrap-Up (~5 min.)	Self-reflection on lesson concepts (<i>SE p. 111</i>)	
	Resources	Glossary		Materials		Additional Preparation
		<u>Key Terms:</u> - Venn diagram - Negation <u>Additional Glossary:</u> - Converse - Inverse - Contrapositive		N/A		For 2.6.2 Guided Instruction, consider displaying overlapping Venn diagrams and concentric Venn diagrams. Then discuss which type of Venn diagram would be used to represent the statements from question 4 for additional examples.

 Teacher Tips	Common Misconceptions • Students may write the inverse statement as the converse or vice-versa. • Students may forget to switch the hypothesis, p, and the conclusion, q, when writing a contrapositive statement. • Students may mistake a negation for another statement type, such as converse. • Students may assume that statements written in conditional form are always true.	Things to Consider • This lesson requires the students to have a good understanding of the concepts in the prior three lessons. Consider using key moments in this lesson to provide additional support for students who need a review of that material.
	Literacy and Language Routines Stronger and Clearer Each Time 2.6.2 Guided Instruction introduces students to “all” and “not” statements. Provide students with your thinking as you help them move through the activity. This will allow students to model their thinking after yours as they are learning the new concepts.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.5.1: Patterns and Structure In 2.6.2 Guided Instruction, students are introduced to “all” and “not” statements. They will connect the structure of conditional statements and truth values to better understand the formal definition of an “all” statement and a negation.
	 Differentiate Instruction	
Lesson Notes:		

2.6.1 Warm-Up: Bell Work

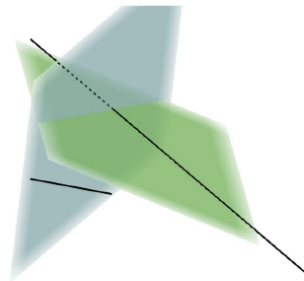
Component Overview: Students will complete a Bell Work covering the previous lesson. Students will write the converse, inverse, and contrapositive statements for a given conditional and determine their truth values. Students will also determine if the statement is a biconditional, writing the biconditional if it is or justifying why not if it is not.



2.6.1 Warm-Up: Bell Work

- Complete the table for the conditional statement.

If two lines are skew, then the lines do not lie on the same plane.



Skew lines are defined in Unit 1 Lesson 1 (G.1.1) if students need to reference the definition.

Use the data collected from this Bell Work to determine remediation needs and focused support during this lesson.

Relationship to Conditional	Statement	True or False
Converse	<i>If two lines do not lie on the same plane, then the lines are skew.</i>	<i>False</i>
Inverse	<i>If two lines are not skew, then the lines lie on the same plane.</i>	<i>False</i>
Contrapositive	<i>If two lines lie on the same plane, then the lines are not skew.</i>	<i>True</i>

- If the conditional statement is biconditional, then write it as a biconditional. If the conditional statement is not biconditional, then explain why not.
The statement is not biconditional because the converse is false. (Skew lines do not intersect and are not parallel)

2.6.2 Guided Instruction: “All” and “Not” Statements

Component Overview: Students will receive formal instruction on “all” statements and negation.



2.6.2 Guided Instruction: “All” and “Not” Statements

Mathematical statements may not be conditional or biconditional. Mathematical statements could be written using “all.” The statement must be universally true to use the word “all.”

1. Discuss with your partner if the statement, “All quadrilaterals have four sides” is universally true. Summarize your discussion.
Answers will vary. Yes, the definition of quadrilateral is that it has four sides.
2. Rewrite “All quadrilaterals have four sides” as an if-then statement.
If a shape is a quadrilateral, then it has 4 sides.
3. Discuss with your partner why the statement “Some quadrilaterals have four sides” is not true. Summarize your discussion.
Answers will vary. “Some” means there may be at least one quadrilateral that does not have only four sides. That is not true. There are no quadrilaterals that have a different number of sides.
4. Determine if each statement is true.

Statement	True or False
All dogs are Labradors.	<i>False</i>
All polygons have congruent sides.	<i>False</i>
All cats are mammals.	<i>True</i>

5. Rewrite each false statement using “some.”
Some dogs are Labradors.
Some polygons have congruent sides.



After questions 1 and 3, create opportunities for partner pairs to share out their answers along with rationale for everyone in the class to hear. Consider using an agree/disagree protocol in order to allow students to have a more productive dialogue with rationale and evidence.



How does changing a statement by adding the word “some” potentially change its truth value?

2.6.2 Guided Instruction: “All” and “Not” Statements (continued)

6. Write each statement as an if-then statement.

Statement	If-Then
All dogs are Labradors.	<i>If there are dogs, then they are Labradors.</i>
All polygons have congruent sides.	<i>If a figure is a polygon, then it has congruent sides.</i>
All cats are mammals.	<i>If an animal is a cat, then it is a mammal.</i>

It may be helpful to understand “all” statements using Venn diagrams.

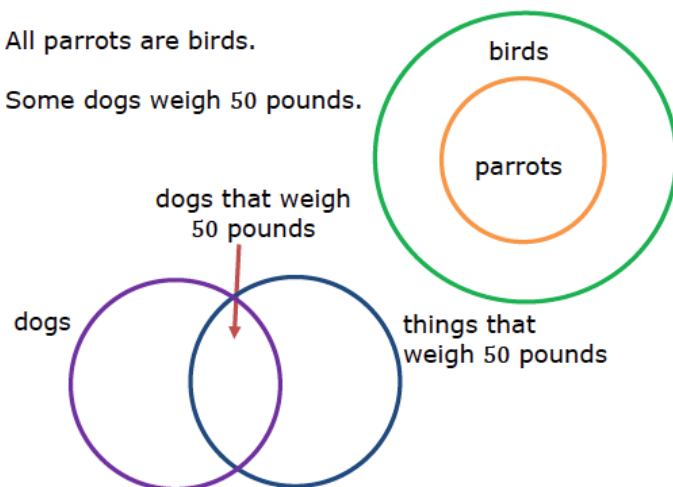


A **Venn diagram** is a diagram representing mathematical or logical sets pictorially as circles. Common elements of the sets are within the overlap of the circles.

7. The Venn diagram and a statement are shown.

All parrots are birds.

Some dogs weigh 50 pounds.



Consider using an anchor chart to help provide a visual example of the Venn diagram concept as it appears in this lesson. This way, students are able to refer back to it for future lessons when needed.



Students have learned about Venn diagrams in earlier grades and other content areas, but it may be beneficial to review the basic structure and function of a Venn diagram as part of this activity using an example that your specific students can easily relate to.

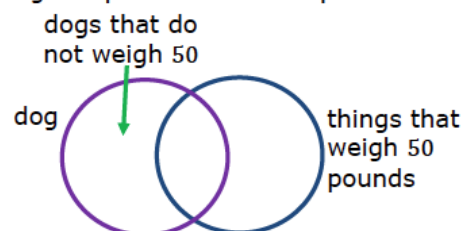
For more practice, represent the statements from question 4 using Venn diagrams.

2.6.2 Guided Instruction: “All” and “Not” Statements (continued)

Discuss with your partner how the Venn diagrams help to show the difference between “all” and “some.” Summarize your discussion.

Answers will vary. If one circle (set) is drawn entirely inside of another, then “all” elements inside that circle (set) also belong to the larger circle (set). If two circles (sets) overlap, then “some” of the elements may belong in the overlapped part of both circles (sets) and “some” of the elements may belong in only one circle (set) but not the other, which is the nonoverlapped parts.

8. The Venn diagram below shows where dogs that do not weigh 50 pounds would be placed.



Write a statement that matches the Venn diagram.

Some dogs do not weigh 50 pounds.



The **negation** of a statement is the opposite of the statement. If the statement is true, then the negation must be false.

9. Use the definition of negation to write the negation of each statement.
- Today is Wednesday. *Today is not Wednesday.*
 - Seo-jun is not wearing blue. *Seo-jun is wearing blue.*



The term “negation” may not be new for students, as it could have been used in other courses. It will be important for students to understand what is intended by the definition of negation in context of this concept.

Students may also confuse the negation of the statement as being synonymous with an inverse or converse statement. Consider front-loading this information when reviewing the terminology.

2.6.2 Guided Instruction: “All” and “Not” Statements (continued)

10. Rolando and Rhianna both wrote the negation of the statement, “ $\sqrt{5}$ is an irrational number.”

Rolando	Rhianna
$\sqrt{5}$ is not an irrational number.	$\sqrt{5}$ is a rational number.

- a. Discuss with your partner both negations. Write your observations and questions for the class.
Answers will vary. We discussed that each statement is the same. If a number is not an irrational number, then it is rational. So, to say a number is not irrational, is the same as saying that it is rational.
- b. Based on the class discussion, which student has written a correct negation?
Both answers are mathematically equivalent.



Students learned about rational vs irrational numbers in a prior course. Depending on their middle school math progression, their exposure to the concept may be limited. Consider reviewing the concept between the two as part of reviewing question 10. Note that for Rhianna to be correct, there is an assumption that she is operating within the real-number system.

2.6.3 Guided Practice: “All” Statements and Negation

Component Overview: Students will engage in a structured practice opportunity with “all” statements and negation. Consider the use of a gradual release model when facilitating this activity.



2.6.3 Guided Practice: “All” Statements and Negation

1. Determine if each statement is true.

Statement	True or False
All congruent angles have the same measure.	<i>True</i>
All odd numbers are prime.	<i>False</i>
All points that lie in a plane are collinear.	<i>False</i>

2. Rewrite each false statement using “some.”

Some odd numbers are prime.

Some points that lie in a plane are collinear.

3. Write each statement as an if-then statement.

Statement	If-Then
All congruent angles have the same measure.	<i>If angles are congruent, then they have the same measure.</i>
All odd numbers are prime.	<i>If a number is odd, then the number is prime.</i>
All points that lie on a plane are collinear.	<i>If points lie on the same plane, then the points are collinear.</i>

4. Write the negation of each statement.

Statement	Negation
$\triangle DTM$ is a right triangle.	<i>$\triangle DTM$ is not a right triangle.</i>
$m\angle TYW + m\angle WYK = 136^\circ$	<i>$m\angle TYW + m\angle WYK \neq 136^\circ$</i>
$\angle RWF$ and $\angle HYV$ are complementary angles.	<i>$\angle RWF$ and $\angle HYV$ are not complementary angles.</i>



With this activity, it is important to note that questions 1-3 are a continuation of the same scenario and are not three unique problems for students. If students need additional scaffolding with this concept, consider modeling one of the statements in question 1.



Question 4 is a unique question that is not part of the learning series of questions 1-3. Consider releasing students to work with a partner or small group to complete question 4.

To add physical movement, consider having students or student pairs find an additional student or pair to check their answers with.

2.6.4 Wrap-Up: Reflection

Component Overview: Students will complete a self-reflection where they rate their level of understanding of concepts from this lesson. This activity should be done independently to provide an insight into each student's level of confidence with the content of this lesson.



2.6.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept below.

Answers will vary.



I need help.



I got it.



I could help someone.

I can write the converse of a mathematical statement.	  
I can write the inverse of a mathematical statement.	  
I can write the contrapositive of a mathematical statement.	  
I can determine if a statement is biconditional.	  
I can write and interpret an "all" statement.	  
I can write and interpret a negation.	  



The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.